

FEDERICO BRUZZONE • CURRICULUM VITAE

PERSONAL INFORMATION

Born in Magenta (MI), Italy on 7th of **March 2000**
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CONTACT INFORMATION

github.com/FedericoBruzzone
[@federicobruzzone](https://twitter.com/federicobruzzone)
[in/federico-bruzzone](https://www.linkedin.com/in/federico-bruzzone)
[@fedebruzzone7](https://x.com/fedebruzzone7)
[u/FedericoBruzzone](https://www.youtube.com/u/FedericoBruzzone)

Presently a PhD Candidate in Computer Science. I have a strong passion for **Programming Languages** and **Compilers**, and I'm currently diving into **AI (ML/DL) compilers**. For more information, just visit my personal [webpage](#).

SCIENTIFIC PUBLICATIONS

- [1] F. Bruzzone, W. Cazzola, M. Brancaleoni, and D. Pellegrino, "Sink or SWIM: Tackling Real-Time ASR at Scale," *IEEE Transactions on Audio, Speech and Language Processing*, no. , pp. 1–11, 2026, doi: [10.1109/TASLPRO.2026.3675780](https://doi.org/10.1109/TASLPRO.2026.3675780). **Journal Ranked Q1 on Scimago** — [bib] [pdf] [IEEE] [arXiv]
[2] F. Bruzzone, W. Cazzola, and L. Favalli, "Code Less to Code More: Streamlining Language Server Protocol and Type System Development for Language Families," *Journal of Systems and Software*, p. 112554, Sept. 2025, doi: [10.1016/j.jss.2025.112554](https://doi.org/10.1016/j.jss.2025.112554). **Journal Ranked Q1 on Scimago** — [bib] [pdf] [SpringerLink] [arXiv]

PREPRINTS PUBLICATIONS

- [3] F. Bruzzone, W. Cazzola, and L. Favalli, "Generalized Software Product Line Extraction." [Online]. Available: <https://arxiv.org/abs/2605.28989> — [bib] [pdf] [arXiv]
[4] F. Bruzzone and W. Cazzola, "Meta-Monomorphizing Specializations." [Online]. Available: <https://arxiv.org/abs/2602.12973> — [bib] [pdf] [arXiv]
[5] F. Bruzzone, W. Cazzola, and L. Favalli, "From Separate Compilation to Sound Language Composition." [Online]. Available: <https://arxiv.org/abs/2602.03777> — [bib] [pdf] [arXiv]
[6] F. Bruzzone, W. Cazzola, M. Brancaleoni, and D. Pellegrino, "Sink or SWIM: Tackling Real-Time ASR at Scale." [Online]. Available: <https://arxiv.org/abs/2601.17097> — [bib] [pdf] [arXiv] **[PUBLISHED]**
[7] F. Bruzzone, W. Cazzola, and L. Favini, "Prioritizing Configuration Relevance via Compiler-Based Refined Feature Ranking." [Online]. Available: <https://arxiv.org/abs/2601.16008> — [bib] [pdf] [arXiv]
[8] F. Bruzzone, W. Cazzola, and L. Favalli, "Code Less to Code More: Streamlining Language Server Protocol and Type System Development for Language Families," *Journal of Systems and Software*, p. 112554, Sept. 2025, doi: [10.1016/j.jss.2025.112554](https://doi.org/10.1016/j.jss.2025.112554). — [bib] [pdf] [arXiv] **[PUBLISHED]**

EDUCATION

2024-Present	PhD Candidate in Computer Science at the ADAPT Lab , University of Milan
2022-2024	MSc in Computer Science at University of Milan (<i>110/110 cum laude</i> , W. Cazzola, 15/07/2024) Thesis: " Toward a Modular Approach for Type Systems and LSP Generation ".
2019-2022	BSc in Musical Computer Science at University of Milan (13/10/2022)
2011-2019	Piano and Music Composition at I.S.S.M. Novara Conservatory
2014-2019	Diploma in Computer Science and Telecommunications at E. Alessandrini

OPEN SOURCE CONTRIBUTIONS (SELECTED)

2026	The Eter Programming Language , a new programming language leveraging the MLIR/IREE ecosystem with tensors support.
2026	MLIR/LLVM: [mlir][ArmSVE] Fix comment inconsistency for <code>pack_lhs</code> in <code>ArmSVE/pack-unpack-mmt4d.mlir</code> (NFC)
2026	MLIR/LLVM: [mlir][ArmSME] fix f64 scalable matmul crashes in <code>VectorLegalizationPass</code>
2026	MLIR/LLVM: [mlir][VectorToLLVM] add opt-in <code>enable-gep-inbounds-nuw</code> pass flag for <code>vector.load/store</code>
2026	MLIR/LLVM: [mlir][MemRefToLLVM] fix incorrect <code>nw</code> on <code>GEP/mul</code> when lowering <code>memref.load/store</code> with negative strides
2026	MLIR/LLVM: [NFC][mlir][linalg] add <code>toContractionDimensions</code> for healthy code reuse
2026	MLIR/LLVM: [mlir][vector] extend <code>createReadOrMaskedRead/createWriteOrMaskedWrite</code> with permutation map support

2026	MLIR/LLVM: [mlir][affine] emit <code>in_bounds</code> on <code>transfer_read/write</code> when statically provable in <code>affine-super-vectorize</code>
2026	IREE: [LLVMCPU] Fix ArmSME requiring classic SVE, breaking SME-only targets (e.g., Apple Silicon)
2026	IREE: [LLVMCPU] Add SME lowering-strategy tests for f64 and unsupported i8 matmuls
From 2026	Maintainer of the LLVM Pass Template, a C++ template project to quickly create new LLVM passes.
From 2025	Maintainer of the Papers on Compiler Optimizations: Analysis and Transformations, a curated list of scientific publications
From 2025	Maintainer of scribe , a minimalist, opinionated LaTeX document class for technical writing and presentations.
From 2025	Maintainer of the Tide Compiler, an backend-agnostic IR and compiler framework written in Rust.
2025	Generalizing Closeness centrality to weighted networks using Newman method (issue 1384)
From 2024	Maintainer of the cross-platform tgt project and tdlib-rs TDLib bindings written in Rust.

RESEARCH ACTIVITIES

2024-Present	MUSEMI Session Chair, Speaker and Co-organizer
May 2026	Committee Member at the Int. Conf. on Software Language Engineering (SLE 2026), ACM, B on CORE
Nov 2025	Reviewer for the Journal of Computer Languages (COLA), Elsevier, C on CORE
Jun 2025	Reviewer for the Journal of Software and Systems Modeling (SoSyM), Springer, Q1 on Scimago
Dec 2025	Reviewer for the Eur. Conf. on Object Oriented Programming (ECOOP 2026), ACM, A on CORE
2-6 Jun 2025	Participant at <Programming> 2025 conference
Apr-Jun 2025	Committee Member at the Int. Conf. on Software Language Engineering (SLE 2025), ACM, B on CORE
Mar 2025	Reviewer for the Journal of Systems and Software (JSS), Elsevier, Q1 on Scimago
2-7 Sep 2024	Student Volunteer at the Int. Conf. on Functional Programming (ICFP 2024)
2022-2024	Research Internship at ADAPT Lab, working on modular type systems and LSP generation for Neverlang
2021-2022	Research Internship at LIM Lab, working on the IEEE 1599 standard

TEACHING ACTIVITIES

THESIS SUPERVISION

14/07/2026	M. Visconti, Towards Neverlang3: A Generative Rust Approach for Resilient LL Parser Generators , MSc, 102
23/02/2026	D. Cerato, Employing Metaprogramming: A Macro-Driven Strategy to Overcome Compiler-Imposed Limitations on Specialization , BSc, 103
10/04/2025	D. Pellegrino, Scalable Multi-client Real-time Whisper , BSc, 96
09/04/2025	A. Longoni, GUIDE: Graphical User Interface Development Environment , MSc, 110L
09/04/2025	L. Albani, New Generalized Protocol For Software Product Line Extraction And Configuration , MSc, 110L
09/04/2025	G. Esposito, Fr3D: A Framework for DAP-compatible DSL-oriented Debugging , MSc, 110L
24/02/2025	L. Favini, RustyEx: Instrumenting rustc to Extract Feature Dependency Graphs , BSc, 102

GRADUATE COURSES

2025-2026	Mathematical Logic (Art. 45), BSc in CS, University of Milan, S. Aguzzoli
2024-2025	Programming 1 (Art. 45), BSc in CS, University of Milan, L. Capra (coordinator W. Cazzola)
2024-2025	Mathematical Logic (Art. 45), BSc in CS, University of Milan, S. Aguzzoli
2023-2024	Mathematical Logic, BSc in CS, University of Milan, S. Aguzzoli
2023-2024	Computer Science, BSc in Comm. and Society, University of Milan, A. Momigliano
2023-2024	Programmin 1, BSc in CS, University of Milan, A. Trentini (coordinator P. Boldi)
2023-2024	Programming in Python, MSc in Chemistry, University of Milan, M. Monga

ADDITIONAL ACTIVITIES

2023-Present	Private Tutoring in Computer Science, Mathematics and Physics
Nov 2024	Collaborator at the Bebras Challenge , ALaDDIn Lab
Jan-Jun 2024	Organizer of the BSc Computer Science Laboratories, ALaDDIn Lab
Jan-Jun 2024	Collaborator of the Workshops for schools, ALaDDIn Lab
Nov 2023	Collaborator at the Bebras Challenge , ALaDDIn Lab

TECHNICAL SKILLS

Languages	Rust, C/C++, Python, Go, OCaml, Java, Scala, Kotlin, Erlang, Lua, Dart, PHP, HTML/CSS, SQL, Bash, TeX
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Systems/Tooling	Git, CI/CD, Docker, GDB, Valgrind, Build Systems (e.g., CMake, Cargo, Pip), Cross-language Linking, Static/Dynamic Libraries, and FFI (e.g., C bindgen)
Area of Expertise	Compiler Construction and Optimizations, Programming Languages Design, IR Design and Implementation, Type Systems, Language Support Tools (e.g., LSP), Static Analysis, Rustc Internals, MLIR, LLVM, Parsing Techniques and Parser Generators (e.g., ANTLR)

MUSICAL ACTIVITIES

2021-Present	Sound Engineer and Music Producer , in collaboration with Believe and as a SLAE member.
2006-Present	Pianist and Music Composer , composing original pieces and arrangements for piano and various ensembles
2019-Present	Piano and Music Teacher , teaching piano and music theory and composition to students of all ages and levels

GRANTS AND FELLOWSHIPS

2024	56th Top Github Public Contributor in Italy out of 958
2023-2024	Scholarship for the MSc in Computer Science, awarded by the University of Milan
2020-2024	Scholarship for the BSc in Musical Computer Science, awarded by the University of Milan

TONGUES

Italian	Mother tongue
English	Level CEFR B2 (SLAM at University of Milan)
Spanish	Base (A1-A2)

Milan, 20/07/2026